

Part A

Client Brief SmartCare currently relies on spreadsheets and paper records. Staff experience duplicate bookings, difficulty finding patient information, inconsistent appointment statuses and limited appointment history. The proposed solution is a small and maintainable system for managing patient, practitioner and appointment information.

Part B

In Scope

- Create, view, update and search patient records.
- Create, view and update practitioner records.
- Create, view, update and cancel appointments
- Link each appointment to a patient and practitioner
- Use consistent appointment statuses
- Detect or prevent duplicate appointment bookings
- Detect or prevent duplicate appointment bookings

Provisional Until Confirmed

- Practitioners directly log in to view their schedules.
- Cancelled appointments are retained indefinitely in history.
- Role-based access, audit logging and specific performance targets.

Out of Scope

- Online payments and billing.
- SMS or email reminders
- Facial recognition

StakeHolder	Need	Evidence
Reception Staff	Efficiently find information and create, update, or cancel appointments for patients.	Report states staff problems with booking, information, status, and history.
Practitioners	Schedules and appointment statuses that are accurate.	Brief said practitioners need appointment management
Patient	Personal information, appointments, and history.	Patient information and appointments are managed by the proposed system.
Management	Mainainble system that reduces duplicate bookings and inconsistent records	Stated in brief.

Developers and testers	Testable requirements and maintainable business logic	Necessary delivery stakeholders.
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Part C

FR-01: System shall allow authorised staff to create a patient record. Why: supports management of patient information.

FR-02: system allow authorised staff to view and update a patient record. Why: Supports correction and maintenance of patient information.

FR-03: System shall allow authorised staff to search for a patient using an approved identifier or search field. Why: Directly addresses difficulty finding patient information; the exact search fields are an open question.

FR-04: System shall allow authorised staff to create, view and update practitioner records. Why: The requested solution must manage practitioners.

FR-05: Allow authorised staff to create an appointment linked to one patient and one practitioner. Why: Defines the central appointment capability and its record relationships.

FR-06: System check for a conflicting booking before saving an appointment. Why: Directly addresses duplicate bookings; the definition of a conflict must be confirmed.

FR-07: Shall display an appointment's patient, practitioner, date, time, and current status. Why: Provides the information needed to identify and manage an appointment.

FR-08: The system shall allow authorised staff to change an appointment using the approved appointment status values. Why: Directly addresses inconsistent appointment statuses

FR-09: The system shall allow authorised staff to cancel an appointment without deleting its record. Why: Supports cancellation while preserving history; the retention rule requires confirmation.

Part D

NFR-01: Maintainability: patient, practitioner and appointment business logic shall be separated from the user interface so it can be changed and tested independently. Why: The brief explicitly asks for a maintainable system.

NFR-02: Data integrity: If an appointment doesn't have a legitimate patient, practitioner, date, time, or approved status, the system will refuse it. Why: Prevents inconsistent or incomplete recordings of appointments.

NFR-03: Reliability: The stored record must be accessible even after the programme has been closed and reopened once the system has verified a save. Why? Because it makes persistence testable and does not presume the necessary storage technology.

NFR-04: Usability: A receptionist must be able to find an existing patient and begin making an appointment on their own after receiving the agreed-upon training; the client must confirm the acceptable time and mistake rate. Why? Because it substitutes a testable task with open-ended objectives for vague "easy to use" terminology.

Part E

US-01: As a receptionist, I want to search for a patient, so that I can find the correct patient before making a booking.

US-02: As a receptionist, I want to create an appointment linked to a patient and practitioner, so that the booking details are recorded consistently.

US-03: As a receptionist, I want the system to warn me about a conflicting booking, so that I can avoid duplicates.

US-04: As a receptionist, I want to update an appointment's status, so that staff can see its current state.

US-05: As a receptionist, I want to cancel an appointment without removing its record, so that the appointment can remain in history if the client approves this rule.

Acceptance Criteria

AC-01 Successful patient search

GIVEN an existing patient record with an approved identifier

WHEN an authorised receptionist searches using that identifier

THEN the system displays the matching patient's information.

AC-02 Successful appointment creation

GIVEN a valid patient, practitioner, date, time and approved status, with no conflicting booking

WHEN an authorised receptionist saves the appointment

THEN the system stores the appointment and displays a confirmation.

AC-05 Appointment cancellation

GIVEN an existing active appointment

WHEN an authorised receptionist cancels it

THEN the system changes its status to Cancelled and does not remove the record, subject to the approved retention rule.

Part F

Review prompt Act as a software requirements reviewer. Review the SmartCare requirements for ambiguity, inconsistency, missing clarification questions and testability. Do NOT invent new client requirements. For every suggestion, state whether it is based on evidence or is only a question/assumption requiring validation.

Part G

Define duplicate booking precisely.

Evidence: Brief reports duplicate bookings.

Decision: Accepted

Reason: A conflict rule is needed to make the requirement testable. Verification: Ask which combinations and time overlaps count as conflicts.

Make search performance measurable.

Evidence: Brief reports difficulty finding patient information.

Decision: Modified

Reason: AI cannot choose the response time or dataset size. Verification: Obtain client-approved values and run a timed test.

Retain cancelled appointments forever.

Evidence: Brief only says appointment history is limited.

Decision: Unverified

Reason: The retention period is not stated. Verification: Confirm retention and deletion rules with the client.

Add SMS reminders.

Evidence: No supporting evidence.

Decision: Rejected

Reason: Not requested; adds integration scope. Verification: Include only if the client approves a scope change

Add facial recognition login.

Evidence: No supporting evidence.

Decision: Rejected

Reason: Authentication is unspecified; biometrics add complexity and risk. Verification: Ask for security requirements without assuming biometrics.

Assumptions and Open Questions

- Which user roles will operate the system, and what may each role view or change?

- Which patient fields, practitioner fields and appointment fields are mandatory?
- Which patient identifiers or search fields must be supported?
- What counts as a duplicate or conflicting appointment: same practitioner, same patient, same time, or overlapping time?
- What are the approved appointment statuses and allowed status changes?
- Should cancelled appointments remain in history, and for how long?
- Do practitioners need direct access to their schedules?
- What response time and dataset size define acceptable search performance?
- Are login, role-based access, audit logging, backup or privacy controls required?
- Are existing spreadsheet and paper records to be imported?

Reflection

The AI review noticed that several requirements sounded reasonable but were not measurable. For example, saying that patient search should be fast does not identify an acceptable response time or the size of the dataset used for testing. It also identified missing clarification questions about duplicate bookings, appointment statuses, permissions and record retention. These points improved the requirements because they made the uncertain areas visible instead of hiding them behind vague wording.

However, by recommending features like online shopping, face recognition, SMS reminders, or medical recommendations, AI did some overreaching with these features. Since none of these features are supported by the SmartCare brief, they were either rejected or left out of the scope. The duplicate-booking condition was changed after review. Instead of merely stating that the system should avoid duplication, it now stipulates that the system must look for a conflicting booking before storing. Requirements need evidence because the system should solve the client's actual problem, not implement features merely because they seem useful. Evidence also allows each requirement to be traced, justified, tested and approved.